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REPLICATION STUDY

# Revisiting US food aid and civil conflict

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**Abstract**
Nunn and Qian (Am Econ Rev 104:1630–1666, 2014) find that an increase in US food aid increases the incidence of civil conflicts. However, their identification strategy (based on shift-share instrumentation) is invalid because the shares used to construct the shift-share instrument do not add up to one. Even when the shifts are exogenous, countries with larger shares systematically have higher values of the instrument, introducing bias if these countries have different unobservables that affect conflict. The exogeneity of the instrument can be restored by adding a sum of shares control interacted with year fixed effects. In Nunn and Qian (Am Econ Rev 104:1630–1666, 2014), the sum of shares control is time-invariant and country-specific, naturally giving rise to a common factor model representation. We replicate Nunn and Qian (Am Econ Rev 104:1630–1666, 2014) using common factor models. We find evidence that US food aid reduces the incidence of conflicts. Our baseline estimates suggest a 10% increase in US food aid would result in a 0.3 percentage point decrease in the incidence of conflicts (or a 1.5% decrease at the sample mean). Our results are robust to first-differenced and dynamic model specifications. We also find that this conflict-reducing effect does not disappear after the Cold War. We also find that US food aid decreases the incidence of small-scale conflicts, the incidence of civil conflicts, and the onset of civil conflicts.

**Keywords** US food aid · Civil conflict · Common factor model · Shift-share

**JEL Classification** D74 · F35 · O17 · O19 · Q11 · Q18

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## 1 Introduction

This paper revisits the seminal findings of Nunn and Qian (2014, *American Economic Review*, 104(6): 1630–1666; hereafter NQ), who find that increases in US food aid raise both the incidence and duration[1] of civil conflicts. Their baseline estimate implies that increasing US food aid by 1,000 metric tons increases the incidence of civil conflicts by 0.3 percentage point in recipient countries.

NQ's findings have attracted substantial scholarly and policy attention, having been cited more than 2200 times as of late 2025 (according to Google Scholar). Additionally, US policy documents implicitly relay their findings into policy guidance (Central Intelligence Agency 2014; Global Food Security Act 2016). Given that the USA has historically been the world's largest food aid donor, these findings carry significant implications for the design and implementation of humanitarian policy and global security.[2]

However, several researchers have questioned the robustness of NQ's results. An early study by USAID (2014) suggests that their findings may be sensitive to the exclusion of unreliable data (1971–1974) or model specification. Recently, Sheppard (2021) shows that NQ's results may be sensitive to the sample period (pre/post-Cold War).[3] Furthermore, the comparison between their ordinary least squares (OLS) and two-stage least squares (2SLS) estimates does not fit the expected reverse causality between conflict and aid (Christian and Barrett 2017).[4] More fundamentally, Christian and Barrett (2017, 2023) suggest that NQ's estimates are unreliable because their identification strategy is flawed.

NQ's approach relies on 2SLS estimations with Bartik-shift or shift-share (SS) instrumental variable (IV). In particular, the SS instrument for food (wheat) aid is the interaction of last year's US wheat production ("shift") and the frequency that a

[1] Note that this result on the duration of conflicts does not come from their main identification strategy, but from the use of a survival model combined with a control function approach, and is therefore beyond the scope of this replication.

[2] Since January 2025, the US food aid program has undergone sweeping changes. The Trump administration implemented a series of measures that substantially reduced USAID's operations (i.e., furloughing staff, cutting budgets, canceling grants, reducing World Food Programme—WFP—contributions, and attempting to eliminate core initiatives such as Food for Peace or McGovern-Dole) and have worked toward major restructuring (even elimination) of USAID functions, with some of those efforts currently subject to legal challenge. As a result, numerous food aid projects have been halted, delayed, or canceled, marking a sharp retreat from global humanitarian engagement and drawing strong criticism from aid organizations and the WFP. In light of these recent political developments that have profoundly reshaped the structure and priorities of food aid and USAID, we believe the relevance of our study is greater than ever. The widely cited work of Nunn and Qian (2014) has frequently been interpreted as providing empirical justification for scaling back or discontinuing US food aid (e.g., Edwards et al. 2023). Reassessing their conclusions within a valid econometric framework is therefore critical to inform current debates on the future of humanitarian assistance. In addition, given that the USA holds presidential elections every four years, a change in administration in the next years could swiftly reverse these restrictive measures, underscoring the cyclical and politically contingent nature of US food aid policy.

[3] Especially, he finds that US food aid increases both conflict duration and the likelihood of conflict onset during the Cold War (1971–1991), but has no effect on conflict duration or onset after the Cold War (1991–2006).

[4] Chu et al. (2017) provide a test of the parametric modeling form used in Nunn and Qian (2014) and confirm their findings using a nonparametric modeling approach.

country receives any US food aid ("share").[5] Baseline regressions also include a set of country fixed effects, region-year fixed effects, and a series of year fixed effects interacted with other controls.

Christian and Barrett (2023; hereafter CB) argue that NQ's estimates are spurious because of uncontrolled trends in conflict, food aid, and the interacted instrument. Indeed, they explain that the identification can only be causal if all the included fixed effects capture all of the country-specific and nonlinear trend effects other than aid (Christian and Barrett 2017). In the face of this problem, CB (2023) offer a suggestion: to first-difference the variables with trends (i.e., conflict, aid, and instrument). Using these first-differenced variables, they find no effect of US food aid on civil conflict.[6]

Yet, CB's approach raises two concerns. First, the use of first-differenced variables modifies the structural model used in NQ (2014). To put it simply, while this solution helps examining the year-to-year changes in food aid on conflict changes, one is still left wondering what the effect of US food aid on conflict is. Second, diagnostic tests for cross-sectional dependence (CSD) indicate both the dependent variable (i.e., conflict), the endogenous regressor (i.e., US wheat aid), and the SS instrument, exhibit (semi-)strong CSD in NQ (see Table 8 in Appendix). This degree of dependence can produce misleading inference and even bias when conventional panel data estimators are applied[7] (e.g., Chudik and Pesaran 2015). Likewise, CSD tests confirm that CB's strategy fails to remove the (semi-)strong cross-sectional dependence affecting the instrument in NQ's set up,[8] potentially leading to bias in their estimates.

While we agree with CB (2023) that NQ's identification is problematic, we do not believe their solution (first-differencing) corrects the main source of bias in NQ. NQ's approach relies on a specific case of SSIV in that aid recipient countries are all exposed to a single shock every year, though to different degrees. The challenge behind any SSIV approach lies in the fact that the IV leverages two sources of variation, i.e., shift and share, and it is unclear what assumptions with respect to both are necessary to guarantee instrument exogeneity (e.g., Goldsmith-Pinkham et al. 2020; Borusyak et al. 2022, 2025; Jaeger et al. 2018; Adão et al,. 2019; Apfel 2024). Several recent studies have sought to address this issue, and their approaches can be broadly divided into two strands. The first focuses on the exogeneity of shares, demonstrating that when all shares are valid, the shift–share IV estimator remains consistent (Goldsmith-Pinkham et al. 2020). The second emphasizes the exogeneity of shifts, showing that identification can instead arise from exogenous shifts themselves (Borusyak et al. 2022). In NQ, identification should come from exogenous shifts (given that shares are likely endogenous). Given their panel data, with many periods, exogeneity of the

[5] Let's note that the SSIV is the result of interacting a country-specific time-invariant share and purely time series exogenous shifts. This particular design will have important one key implication as we later explain.

[6] In their paper, they only show 2SLS results using the shift as a simple instrument and find negative but statistically insignificant estimates. We provide 2SLS results using the SS instrument and the first-differenced variables later in the paper. The key estimates remain positive though they are statistically insignificant.

[7] This eventually depends on the extent of the cross-sectional dependence and on whether the source generating the cross-sectional dependence (e.g., unobserved common shock) is correlated with regressors.

[8] It does, however, remove the issue with respect to the dependent variable, while we find evidence of weak CSD for the first-differenced aid variable.

time series variation in the shifts may be sufficient for consistent estimation (Borusyak et al. 2022).

However, in NQ, the shares used to construct the SS instrument do not add up to one ("incomplete shares"). Even when the shifts are exogenous, countries with larger shares systematically have higher values of the SS instrument, introducing bias if these countries have different unobservables that affect conflict (Borusyak et al. 2022). Borusyak et al. (2022) demonstrate that including an additional control, i.e., the sum of shares interacted with time fixed effects, ensures instrument exogeneity and avoids bias. Neither NQ nor subsequent studies (i.e., CB, 2023; Shepard, 2021) address this issue.

Importantly, in NQ, the sum of shares control is time-invariant and country-specific, naturally giving rise to a common factor model representation, in which the sum of shares control acts as factor loadings and year fixed effects represent common factors. Correcting for incomplete shares implicitly rejoins another aspect of CB's critique. NQ's approach relies on region-year fixed effects to control for unobserved time-varying heterogeneity. This assumes that common shocks have the same effect on all countries within a region of the world.[9] Similar restrictive ways to model time dependence in the conflict literature have produced biased estimates and misleading policy guidance (e.g., Buhaug 2010; Couttenier and Soubeyran 2014; Mary 2022a, 2022b).

In reality, global factors, such as fluctuations in commodity prices, trade policy shifts, or climatic events (e.g., El Niño–Southern Oscillation), may affect recipient countries with varying intensity depending on their economic structures or agroecological characteristics. This heterogeneity generates correlated residuals across countries, violating the standard assumption of cross-sectional independence (as discussed earlier) and leading to potentially biased estimates in NQ.

By contrast, common factor models flexibly allow capturing unobserved time-varying heterogeneity, such as country-specific nonlinear trends, and solve the problems raised in CB (2023). These models relax the assumption that common shocks have the same impact on countries (within a region), addressing the presence of cross-sectional dependence in NQ and CB by explicitly representing unobserved shocks or global factors by a common factor structure, where the error term includes unobserved time-varying factors with heterogeneous unit-specific loadings (e.g., Pesaran 2006; Bai 2009; Chudik and Pesaran 2015).

The literature on common factor models offers several alternatives to estimate panel data models, including the common correlated estimator (Pesaran 2006; Harding and Lamarche 2011 for IV setting; Chudlik and Pesaran 2015 for a dynamic model), the principal components estimator (Kapetanios and Pesaran 2005), the interactive principal components (IPC) approach (Bai 2009), its extensions (Moon and Wiedner, 2015; 2017), and the general two-stage IV (G2SIV) approach (Norkute et al. 2021; Cui et al. 2022). We adopt the latter because it combines the attractive features of both the CCE and the IPC approaches (Kripfganz and Safaridis, 2021). While the G2SIV approach is robust and computationally inexpensive, it also avoids the incidental parameter bias

[9] NQ use the following regions: East Asia & Pacific, Europe & Central Asia, Latin America & Caribbean, Middle East & North Africa, South Asia, Sub-Saharan Africa.

(and thus does not require bias correction), allows for endogenous regressors and/or dynamic models, and deals with unbalanced panels (Kripfganz and Safaridis, 2021; Norkute et al. 2021).

Our results contrast with NQ's conclusions. Using G2SIV estimations, we find that US food aid *reduces* the incidence of conflicts. Our baseline estimates imply that a 1,000 MT increase in US wheat aid would decrease the incidence of conflicts by 0.1 percentage point. At the sample mean, a 10% increase in US food aid would result in a 0.3 percentage point decrease in the incidence of conflicts (or a 1.5% decrease). We also find that US food aid decreases the incidence of civil (or intra-state) conflicts, the incidence of small-scale conflicts, and the onset of new intra-state conflicts. Lastly, we find no effect of US food aid on inter-state conflict.

The remainder of the paper is as follows. Section 2 presents the empirical strategy. Section 3 discusses the estimation results. Section 4 concludes.

## 2 Empirical strategy

### 2.1 Nunn and Qian's approach

To examine the impact of US food aid shipments on conflict in recipient countries, Nunn and Qian (2014) use the following 2SLS modeling approach:

$$C_{it} = \beta F_{it} + \mathrm{X}_{it}\Gamma + \varphi_{rt} + \psi_i + \nu_{it} \tag{1}$$

$$F_{it} = \alpha\left(P_{t-1} * \overline{D}_i\right) + \mathrm{X}_{it}\Gamma + \varphi_{rt} + \psi_i + \varepsilon_{it} \tag{2}$$

where Eq. (1) is the second stage of the 2SLS system and Eq. (2) is the first stage.

$C_{it}$ is a dummy variable that equals one if there is a conflict in country $i$ in year $t$; $F_{it}$ is the quantity of wheat aid shipped from the US to recipient country $i$ in year $t$. $\mathrm{X}_{it}$ is a vector of covariates, namely a set of year fixed effects interacted with several variables, i.e., average recipient cereal production and imports, average US economic, and military aid inflows; monthly recipient temperature and rainfall; and a set of interactions between $\overline{D}_i$ and, respectively, US real per capita GDP, a dummy variable for whether the president is a Democrat, and real oil prices. $\varphi_{rt}$ is set of region-year fixed effects[10] (with $r$ as an index for regions) and $\psi_i$ is country individual effects.

The Bartik-shift or SS instrument $\left(P_{t-1} * \overline{D}_i\right)$ for wheat aid is the amount of US wheat production in the previous year $P_{t-1}$ interacted by the country's propensity to receive food aid from the US $\overline{D}_i$. This is calculated as the fraction of years over the sample period (1971–2006) that a country receives any US food aid. In additional analyses, they employ $P_{t-1}$ as an un-interacted instrument.

Using a sample of 125 countries, they find that a 1000 metric ton increase in US wheat aid increases the incidence of conflicts by 0.3 percentage point. They also show that this overall conflict increase is solely due to an increase in intra-state conflicts as

[10] In the model using the un-interacted model, region-year fixed effects are replaced by region-specific time trends.

they find no effect of US wheat aid on inter-state conflicts. Last, US wheat aid does not seem to affect the onset of new conflicts.

### 2.2 Discussion of shift-share instrument

In the most general form, SSIV are a weighted average of a common set of shifts (or shocks), with weights reflecting heterogeneous exposure to shocks. The shifts vary at a different level of analysis than the one of units. While such instruments have been increasingly used in literature in the last three decades (e.g., Bartik 1991; Blanchard and Katz 1992; Autor et al. 2013), their claim for validity is not evident (Borusyak et al. 2025).

A recent literature has tried to clarify the assumptions about the shifts, or shares, or both, required to ensure identification (e.g., Goldsmith-Pinkham et al. 2020; Borusyak, et al. 2022, 2025; Jaeger et al. 2018; Adão et al. 2019; Apfel 2024). There are two paths to ensure instrument exogeneity: exogenous shifts or exogenous shares. First, Borusyak et al. (2022) demonstrate that the SSIV can be valid when the identification comes from the shifts under certain conditions (namely many random shifts and shares adding up to one). Second, Goldsmith-Pinkham et al. (2020) show that if all shares are valid, the SSIV also is, imposing no implicit assumption on shock exogeneity.

NQ's application naturally aligns with the exogenous shifts approach as the shifts can be regarded as valid instruments. NQ rely on a special case of SSIV in that the shares are time-invariant and the units are exposed to a single shock. The shares are calculated as a recipient country's average of aid exposure over the sample period and are likely endogenous. Still, with the long time series of shifts in NQ, exogeneity of shifts may be sufficient for consistent estimation, as long as the shifts have weak serial dependence,[11] even if the number of shocks is small (Borusyak et al. 2025).

However, there is another issue with NQ's strategy. The shares do not add up to one. Even with randomly assigned shifts, recipient countries with higher shares will typically get higher values of the SSIV. This means the SS instrument is correlated with the shares which in turn may be correlated with the error term, explaining the bias in NQ (Borusyak et al. 2022). There is an easy fix to this issue: Adding the sum of shares control interacted with time fixed effects will restore instrument exogeneity and avoid bias[12] (Borusyak et al. 2022).

### 2.3 Common factors approach

In NQ, the sum of shares control is time-invariant and country-specific, naturally giving rise to a common factor model representation, in which the sum of shares control acts as factor loadings and year fixed effects represent common factors.[13]

[11] We estimate the autocorrelation coefficient around 0.3.

[12] All controls must also be interacted with shares (Borusyak et al. 2025).

[13] Given that there is only one shift, the sum of shares is actually $\overline{D}_i$.

Accordingly, we modify Eqs. (1) and (2) as follows:

$$C_{it} = \beta F_{it} + \mathrm{H}_{it}\Gamma + \mu_i L_t + \psi_i + \nu_{it} \tag{3}$$

$$F_{it} = \alpha\left(P_{t-1} * \overline{D}_i\right) + \mathrm{H}_{it}\Gamma + \mu_i L_t + \psi_i + \varepsilon_{it} \tag{4}$$

where $L_t$ is a set of common factors and $\mu_i$ is a vector of factor loadings. $\mu_i$ represents the heterogeneous effect of unobservable common shocks on each country. Note that the vector H of covariates is different than in Eqs. (1) and (2) because the set of year fixed effects interacted with specific variables becomes redundant. Similarly, region-year fixed effects are not included. The remaining variables are as follows: the interaction of the shares $\overline{D}_i$ and, respectively, US real per capita GDP, a dummy variable for whether the president is a Democrat, and real oil prices; recipients' monthly rainfall and temperature, as well as the interaction of the shares $\overline{D}_i$ with the monthly climate variables.

In general, Eqs. (3) and (4) provide a much more flexible way to address unobserved time-varying heterogeneity than NQ and directly deal with cross-sectional dependence. Indeed, as discussed in the introduction, diagnostic tests for CSD suggest that conflict, US wheat aid, and the SS instrument exhibit semi-strong CSD in NQ. We test the presence of weak cross-sectional dependence using Pesaran (2015) and we estimate the exponent of cross-sectional dependence using Bailey et al. (2016, 2019).[14] This implies the presence of common factors influencing conflict across countries and explicitly validates the need for common factor models.[15] We also test for slope homogeneity and cannot reject the null of homogeneous slopes.

We now discuss several extensions to our baseline model. First, as conflict tends to persist over time, we modify Eq. (3) by including a lagged dependent variable capturing persistence:

$$C_{it} = \pi C_{it-1} + \beta F_{it} + \mathrm{H}_{it}\Gamma + \mu_i L_t + \psi_i + \nu_{it} \tag{5}$$

Equation (5) accounts for serial correlation in conflict and has one major advantage over the static model in Eq. (3) in that it prevents attributing conflict persistence to the residuals thus potentially inflating CSD.

Second, CB (2023) recommend checking the validity of empirical findings by reporting results using first-differences. We therefore present the results of a first-differenced model obtained by first-differencing Eq. (3):

$$C_{it} = \pi C_{it-1} + \beta \Delta F_{it} + \Delta \mathrm{H}_{it}\Gamma + \mu_i \Delta L_t + \Delta \nu_{it} \tag{6}$$

[14] We use the Stata command xtcse2 designed by Ditzen (2021) to implement these tests.

[15] If unobserved common shocks are properly modeled through a common factor structure, CSD is effectively absorbed, and inference is valid (Pesaran 2006; Bai 2009; Chudik & Pesaran 2015). This can be tested ex post using residuals from the estimated common factor models. If some factor structure remains in the residuals, demonstrated by the presence of strong CSD in the residuals, the number of factors may need to be increased (Bai & Ng 2002; Pesaran 2015). In the case of the CCE estimator, the number of lags or cross-sectional averages needs to be increased.

As we move $C_{t-1}$ to the right side of the equation, Eq. (6) is a dynamic model. The $\beta$ estimate from this model captures the impact of US wheat aid changes on civil conflict changes. $\Delta F_{it}$ is instrumented by the first-differenced Bartik-shift instrument ($\Delta P_{t-1} * \overline{D}_i$). Unlike Christian and Barrett (2023), we first difference the entire model to keep consistency with the structural model in Eq. (3).[16]

Third, SSIV in panel data applications bring another challenge related to potential serial correlation in the shifts that could potentially end up in finite sample bias and inference issues (Jaeger et al. 2018; Borusyak et al. 2022). For example, when lagged shifts affect current conflict, the static model in Eq. (3) suffers from an omitted variable bias. Several strategies exist to address this problem, but all rely on modifying the SSIV construction (e.g., first-differencing, see CB, 2023). First, one can estimate a richer model to examine dynamic effects. In practice, NQ's original model could be augmented with a lagged treatment instrumented with a lagged instrument (e.g., Jaeger et al. 2018). This strategy, however, requires more instruments and a correct specification of the dynamic effects. Alternatively, one can modify the instrument by isolating the unpredictable component of the contemporaneous shifts before building the shift-share IV (Borusyak et al. 2022). Given that shifts in NQ are serially correlated of order 1 and follow a quadratic trend, a time series regression controlling for both allows extracting this unpredictable component. We implement this approach in additional analyses.

Last, one might be interested in testing the effect of US food aid on conflict for specific periods like before and after the end of the Cold War (Sheppard 2021). To do so requires the introduction of interaction terms capturing specific periods in the SSIV design. Borusyak et al. (2025) explain that this is similar to having multiple SS variables. Equation (3) can be extended as follows:

$$C_{it} = \beta F_{it} + \mathrm{H}_{it}\Gamma + \gamma F_{it} CW_t + CW_t + \mu_i L_t + \psi_i + \nu_{it} \tag{7}$$

With $CW_t$ as a dummy variable equal to 1 if year > 1991 (to capture the post-Cold War era), 0 otherwise. $\beta$ captures the effect of US food aid on conflict before the end of the Cold War, while $(\beta + \gamma)$ captures the effect of US food aid on conflict after the end of the Cold War. The statistical significance of $\gamma$ indicates that the effect of food aid on conflict is (statistically) different in the first period (pre-1991) than in the second one (post-1991).

### 2.4 Discussion of G2SIV

Several approaches have been proposed to estimate common factor models, e.g., the common correlated estimator (Pesaran 2006; Harding and Lamarche 2011; Chudlik and Pesaran 2015), the principal components approach (Kapetanios and Pesaran 2005), the IPC estimator (Bai (2009), and its extension by Moon and Weidner (2015, 2017).[17]

[16] The country fixed effects are eliminated by first-differencing.

[17] They typically can be classified around two strategies. The first jointly removes interactive effects from both the regressors and the error term in a single step, whereas the second asymptotically purges the interactive effects from the error term only (Cui et al. 2022).

Both the CCE and the IPC approaches have become increasingly popular in recent years (Kripfganz and Safaridis, 2021). At their core, both methods involve least squares and project out the common factors using either cross-sectional averages of observables or principal components analysis. Each approach offers distinct advantages but also presents certain limitations.

As explained earlier, instead, we use the G2SIV approach to estimate common factor models (Norkute et al. 2021; Cui et al. 2022). This approach removes the common factors from the error term and the regressors separately in two stages. In the first stage, the instrumental variable (IV) approach projects out the common factors from exogenous covariates using principal components analysis and constructs instruments from defactored covariates. In the second stage, the entire model is defactored based on factors extracted from the first-stage residuals, and IV regression is implemented again using the same instruments.

The G2SIV approach integrates elements of both the CCE and IPC estimators while offering several notable advantages. First, unlike the CCE and the IPC that suffer from incidental parameter bias[18] and thus require bias correction for valid asymptotic inference, the G2SIV does not require bias correction (Norkute et al. 2021). Second, in contrast to CCE, G2SIV does not rely on the rank condition, since it estimates the common factors through principal component analysis rather than cross-sectional averages (Kripfganz and Sarafidis 2021). Third, the G2SIV objective function is linear in the parameters, making the method robust and computationally inexpensive, unlike the IPC that relies on nonlinear optimization. Finally, the G2SIV framework accommodates endogenous regressors provided suitable external instruments are available, and it can also handle both unbalanced panels and dynamic model specifications[19] (Kripfganz and Sarafidis 2021).

Several choices must be made before the G2SIV approach can be implemented, namely the number of factors and the number of lags. To choose the number of factors, we rely on the criteria from Ahn and Horenstein (2013) and fromOnatski (2010).[20] We find that 8 factors are needed for wheat aid and 1 for the rest of the covariates in our baseline model. As food aid is considered to be endogenous, we use an external instrument, i.e., the interacted instrument from NQ. Alternatively, we can ignore the endogeneity of food aid and instrument food aid using its own lagged values (see Table 13).

To choose the optimal number of lags (and thus instruments), we can rely on the model and moment selection criteria (MMSC), in particular the Bayesian Information Criterion (BIC) and Hannan–Quinn Information Criterion (HQIC) (i.e., select the specification with lowest BIC or HQIC), or the downward testing (DT) procedure[21] (i.e., select the specification with lowest J-stat) from Andrews (1999) and Andrews and Lu (2001). Alternatively, we can minimize the Nagar approximation of the mean

[18] This is because an increasing number of parameters needs to be estimated as T or N grows.

[19] The static G2SIV estimator is asymptotically as efficient as the bias-corrected CCE and IPC estimators (Norkute et al. 2021; Kripfganz and Safaridis, 2021).

[20] We use the Stata command xtivdfreg designed by Kripfganz and Sarafidis (2021) to implement the G2SIV approach, and the command xtnumfac from Ditzen and Reese (2022) to calculate the number of factors. In practice, we use the highest number of factors that either criteria returns.

[21] These criteria and the procedure can be used for the static and dynamic models.

square error derived by Okui (2009) for the dynamic model.[22] At the end, we do not find the number of lags substantially alters the baseline results (see Table 3), so we opt for 4 lags for wheat aid in our baseline results to minimize the loss of observations. Still, we report these criteria whenever relevant. Note they are not always needed, so some tables do not display them for the sake of space. As wheat aid is defactored separately from the rest of the covariates, we use 1 lag to avoid instrument proliferation for all other covariates, as the instrument count is getting closer the number of countries. All other covariates are defactored jointly.

## 3 Results

### 3.1 Baseline results

We begin by replicating Nunn and Qian's (2014) benchmark results. Our estimates in Table 9 (columns 1, 3, and 5) are numerically identical to theirs. Consistent with NQ, we find that US food aid increases the incidence of conflict[23] and of intra-state conflict,[24] while having no effect on inter-state conflict. Further, columns 2, 4, and 6 report pre-trend tests, in which current US food aid is regressed on lagged conflict. The coefficients are positive and statistically significant for all conflict and intra-state conflict outcomes (columns 2 and 4), implying that current US food aid "predicts" past conflict. This is a clear warning sign about NQ's identification strategy.

We then implement Christian and Barrett's (2023) proposed correction by first-differencing conflict, food aid, and the SS instrument. Results are reported in Table 10. The aid estimates for conflict and intra-state conflict, respectively, in columns 1 and 3, remain positive but are no longer statistically significant. The estimate for inter-state conflict (column 5) remains negative and insignificant, as in NQ. In addition, we note that the Kleibergen–Paap statistic (1.51) indicates weak instruments in Table 10. The first-differenced instrument has little explanatory power for the first-differenced food aid variable.

Importantly, the corresponding pre-trend tests in columns 2–4-6 now return statistically insignificant coefficients throughout, indicating that the differenced specification successfully removes the spurious trend patterns that undermine NQ's baseline. However, as previously discussed, the CB approach does not address the incomplete-shares problem, failing to ensure instrument exogeneity, and therefore still delivers biased estimates.

Table 1 implements our empirical strategy and displays our baseline estimation results when we estimate the model in Eq. (3) instrumenting food aid with the SS variable. We control for country and year fixed effects by eliminating them prior to the estimation. We use four lags of the SS instrument as defactored instruments and up to a

[22] Several remarks are noteworthy. First, Okui's method to select the number of lags only apply to dynamic models with strictly exogenous covariates. Second, the BIC and HQIC can be used for static and dynamic models.

[23] This measure of all conflicts includes intra-state and inter-state conflicts, and also, a small number of conflicts labeled by UCDP/PRIO as "extra-systemic" or "internationalized" conflicts (NQ, 2014).

[24] This is the variable capturing civil conflict.

**Table 1** The effect of US food aid on conflict (static model)

| | (1) | (2) | (3) |
|---|---|---|---|
| | Any conflict | Intra-state | Inter-state |
| US wheat aid (1000 MT) | − 0.00121*** | − 0.00102*** | − 0.00005 |
| | (0.00027) | (0.00030) | (0.00004) |
| N | 3,699 | 3,699 | 3,699 |
| Number of countries | 125 | 125 | 125 |
| Number of instruments | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 |
| Number of common factors (aid) | 8 | 8 | 8 |
| Rho | 0.591 | 0.623 | 0.691 |
| J-stat | 53.35 | 53.87 | 45.55 |
| J test, *p*-val | 0.538 | 0.518 | 0.814 |
| BIC | − 825.7 | − 825.2 | − 833.5 |
| HQIC | − 419.9 | − 419.4 | − 427.7 |
| CD stat, residuals | 3.701 | 12.12 | 63.68 |
| CD test, residuals, *p*-val | 0.000 | 0.000 | 0.000 |
| Exponent CSD | 0.532 | 0.634 | 0.456 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component. Country and year fixed effects included. All G2SIV estimates
Sample includes 125 countries between 1971 and 2006. All other controls included (see page 8)

maximum of eight factors. While we have explained the selection criteria for both lags and factors earlier, we test the sensitivity of our baseline results to varying lags and factors in additional analyses. The p-values of the J test statistic across columns indicate that the over-identifying restrictions (instruments) are valid throughout Table 1. The rho values are relatively high throughout the estimations, suggesting that most of the variation in the composite error term is due to the single unobserved factor, conditional on the country, and year fixed effects. Put simply, estimators that do not control for common shocks are likely to be markedly biased.

In contrast with NQ, the estimates in columns 1 and 2 Table 1, respectively, for the incidences of conflict and of intra-state conflict, are negative and statistically significant at 1%. These estimates suggest substantial economic effects. According to column 1, a 1000 MT increase in food aid would result in 0.1 percentage point decrease in conflict incidence. For a country at the sample mean of US wheat aid (2775 MT), a 10% increase in US food aid would approximately result in a 0.3 percentage point decrease in the incidence of conflicts, or a 1.5% decrease at the mean conflict incidence. Our results contrast with NQ and CB. We have explained earlier that both papers use an invalid identification strategy. Correcting for the SS instrument endogeneity via the common factor model gives completely different results.

Our results are, however, very much in line with Mary and Mishra (2020) who find that overall humanitarian food aid decreases civil conflict risks. Their baseline

estimates suggest a 10 percent increase in humanitarian food aid per capita decreases the incidence of civil conflict by about 0.2 percentage point (or by about 0.9% at the mean conflict incidence).

As in Tables 9 and 10, Table 11 reports the pre-trend test results based on our baseline specification. None of the estimated coefficients is statistically significant. This provides reassuring evidence that the common factor approach effectively controls for underlying trends (as does the first-differenced specification proposed by CB) while doing so without requiring any modification to NQ's original structural model (unlike CB).

Table 12 in Appendix shows the estimation results of Eq. (4) using the SS instrument. While these should be seen as a heuristic, rather than the equivalent of a first stage of a 2SLS system, they do seem to suggest that the Bartik-shift instrument is a strong instrument for all columns of Table 12 (see specifications using deeper lags, i.e., 4 to 8, in Table 3). The estimates are all positive and statistically significant. This is in line with the first-stage estimates from NQ.

### 3.2 Alternative specifications

First, Table 13 in Appendix displays estimation results when US food aid is considered to be exogenous. In that case, US food aid is instrumented with its own lagged values. The estimates are negative and statistically significant in columns 1 and 2, respectively, − 0.0003 and − 0.0004, suggesting that US food aid reduces conflict and intra-state conflict. The estimates are, however, much smaller in absolute values than those from Table 1. This fact seems to support the expected reverse causality between conflict and US food aid (Christian and Barrett 2017).

Second, the estimations for the dynamic model described in Eq. (5) are displayed in Table 2 and confirm our earlier findings from Table 1. The aid estimates are all negative and are statistically significant in columns 1 and 2, though the estimates are smaller than those from the static model in Table 1. The estimate from column 1 implies that a 1000 MT increase in food aid would result in 0.05 percentage point decrease in conflict incidence (about half of the static model's estimate in Table 1). The estimate for inter-state conflict is virtually the same than in Table 1 and remains statistically insignificant. The coefficients for the lagged dependent variable are, respectively, 0.76 and 0.28 in columns 2 and 3, and statistically significant at 1%, suggesting high persistence in intra-state conflict and moderate persistence in inter-state conflict.

It is worth noting that the dynamic specification appears to mitigate cross-sectional dependence in the residuals. In Table 3, the estimated CSD exponents indicate only weak correlation across units, whereas Table 1 still shows evidence of strong residual dependence under the static model.

Third, Table 3 examines the sensitivity of our results in Table 1 to varying lags (from 5 to 8 lags). We do not display the results for inter-state conflict for the sake of space and because they confirm the absence of any impacts. Increasing the number of lags of food aid used as defactored instruments does not seem to affect the main pattern of results. All aid estimates are negative and statistically significant throughout

**Table 2** The effect of US food aid on conflict (dynamic model)

| | (1) | (2) | (3) |
|---|---|---|---|
| | Any conflict | Intra-state | Inter-state |
| US wheat aid (1000 MT) | − 0.00051* | − 0.00076*** | − 0.00005 |
| | (0.00031) | (0.00021) | (0.00003) |
| Lagged dependent variable | 0.75727*** | 0.75457*** | 0.28117*** |
| | (0.04661) | (0.02934) | (0.02348) |
| N | 3,571 | 3,571 | 3,571 |
| Number of countries | 124 | 124 | 124 |
| Number of instruments | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 |
| Number of common factors (aid) | 8 | 8 | 8 |
| Rho | 0.321 | 0.350 | 0.640 |
| J-stat | 54.47 | 48.90 | 45.21 |
| J test, *p*-val | 0.456 | 0.671 | 0.797 |
| BIC | − 820.9 | − 826.4 | − 830.1 |
| HQIC | − 417.8 | − 423.4 | − 427.1 |
| Okui MSE (× 1000) | 0.870 | 0.544 | 0.0005 |
| CD stat, residuals | 7.228 | 7.868 | 49.29 |
| CD test, residuals, *p*-val | 0.000 | 0.000 | 0.000 |
| Exponent CSD | 0.480 | 0.297 | 0.463 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component. Country and year fixed effects included. All G2SIV estimates
Sample includes 124 countries between 1971 and 2006. One country is dropped from the sample due to data limitations. All other controls included (see page 8)

Table 3. The range of estimates across columns (0.07–0.1) confirms the existence of substantial economic impacts.

A quick look at the MMSC (both BIC and HQIC) suggests using 8 lags of food aid as defactored instruments while the DT procedure (selecting the lowest J-stat) implies using 4 lags. Given that all estimates are negative and statistically significant in Table 4, using 4 lags as the baseline specification prevents the loss of observations resulting from using deeper lags.

Fourth, we originally set the number of factors to 8 following Ahn and Horenstein (2013) and from Onatski (2010). Table 4 examines the sensitivity of our results to varying factors (from 6 to 10 factors). Again, all aid estimates are negative and statistically significant, suggesting that US food aid reduces the incidence of conflicts and civil conflicts. The range of estimates is in line with those from Table 1 columns 1 and 2.

Tables 14 and 15 reproduce similar tests for the dynamic model. Table 14 varies the number of lags for the dynamic model. All estimates are negative but are not statistically significant for all columns. In particular, as we use deeper lags of the SS

**Table 3** The effect of US food aid on conflict: varying lags (static model)

| | Any conflict | | | | Intra-state | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| US wheat aid (1000 MT) | −0.00104*** | −0.00079** | −0.00072*** | −0.00094*** | −0.00103*** | −0.00073*** | −0.00071*** | −0.00092*** |
| | (0.00030) | (0.00031) | (0.00026) | (0.00017) | (0.00030) | (0.00027) | (0.00026) | (0.00021) |
| N | 3,594 | 3,489 | 3,383 | 3,276 | 3,594 | 3,489 | 3,383 | 3,276 |
| Number of countries | 125 | 125 | 125 | 125 | 125 | 125 | 125 | 125 |
| Number of instruments | 108 | 109 | 110 | 111 | 108 | 109 | 110 | 111 |
| Lags | 5 | 6 | 7 | 8 | 5 | 6 | 7 | 8 |
| Number of common factors (aid) | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| Rho | 0.581 | 0.581 | 0.598 | 0.598 | 0.619 | 0.614 | 0.614 | 0.622 |
| J-stat | 53.52 | 47.65 | 49.11 | 48.99 | 56.19 | 56.15 | 60.36 | 59.10 |
| J test, *p*-val | 0.569 | 0.806 | 0.821 | 0.821 | 0.468 | 0.507 | 0.390 | 0.472 |
| BIC | − 830.7 | − 841.5 | − 844.8 | − 849.5 | − 828 | − 833 | − 833.6 | − 839.4 |
| HQIC | − 423.3 | − 432.8 | − 434.9 | − 438.5 | − 420.7 | − 424.3 | − 423.6 | − 428.4 |
| CD stat, residuals | 4.230 | 8.336 | 8.902 | 11.84 | 14.92 | 18.34 | 16.93 | 16.15 |
| Exponent CSD | 0.526 | 0.615 | 0.608 | 0.771 | 0.518 | 0.590 | 0.546 | 0.586 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component. Country and year fixed effects included. All G2SIV estimates
Sample includes 125 countries between 1971 and 2006. All other controls included (see page 8)

**Table 4** The effect of US food aid on conflict: varying factors (static model)

| | Any conflict | | | | Intra-state | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| US wheat aid (1000 MT) | −0.00104*** | −0.00115*** | −0.00107*** | −0.00109*** | −0.00081*** | −0.00086*** | −0.00107*** | −0.00091*** |
| | (0.00026) | (0.00027) | (0.00025) | (0.00026) | (0.00030) | (0.00032) | (0.00029) | (0.00025) |
| N | 3,699 | 3,699 | 3,699 | 3,699 | 3,699 | 3,699 | 3,699 | 3,699 |
| Number of countries | 125 | 125 | 125 | 125 | 125 | 125 | 125 | 125 |
| Number of instruments | 107 | 107 | 107 | 107 | 107 | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| Number of common factors (aid) | 6 | 7 | 9 | 10 | 6 | 7 | 9 | 10 |
| Rho | 0.585 | 0.589 | 0.586 | 0.587 | 0.616 | 0.618 | 0.625 | 0.620 |
| J test, *p*-val | 0.458 | 0.468 | 0.550 | 0.553 | 0.411 | 0.309 | 0.615 | 0.555 |
| BIC | − 823.6 | − 823.9 | − 826 | − 826.1 | − 822.4 | − 819.4 | − 827.7 | − 826.2 |
| HQIC | − 417.8 | − 418.1 | − 420.2 | − 420.3 | − 416.5 | − 413.5 | − 421.9 | − 420.3 |
| CD stat, residuals | 3.890 | 4.049 | 3.121 | 3.783 | 23.15 | 16.78 | 16.81 | 12.87 |
| Exponent CSD | 0.603 | 0.585 | 0.599 | 0.635 | 0.674 | 0.653 | 0.608 | 0.600 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component. Country and year fixed effects included. All G2SIV estimates
Sample includes 125 countries between 1971 and 2006. All other controls included (see page 8)

**Table 5** The effect of US food aid on conflict: first-differenced model

| | Any conflict | | Intra-state conflict | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| US wheat aid (1000 MT) | − 0.00036# | − 0.00048* | − 0.00069*** | − 0.00080*** |
| | (0.00025) | (0.00026) | (0.00015) | (0.00018) |
| N | 3,447 | 3,323 | 3,447 | 3,323 |
| Number of countries | 124 | 124 | 124 | 124 |
| Number of instruments | 107 | 108 | 107 | 108 |
| Lags | 4 | 5 | 4 | 5 |
| Number of factors (aid) | 8 | 8 | 8 | 8 |
| Rho | 0.330 | 0.337 | 0.375 | 0.374 |
| J test, *p*-val | 0.661 | 0.838 | 0.656 | 0.560 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, # $p < 0.15$
Rho is the fraction of the variance of the error term that is explained by the factor component. Year fixed effects included. All G2SIV estimates
Sample includes 124 countries between 1971 and 2006. All other first-differenced controls included

instrument (i.e., 7 and 8), the aid estimates become statistically insignificant in columns 3–4 and 7–8. Using deeper lags increases the instrument count that is getting closer to the number of countries, which may explain the lack of precision. We also note that the DT procedure indicates that specifications with deeper lags are not preferred compared to the baseline specification (using 4 lags). Table 15 varies the number of factors in the estimation of the dynamic model. All aid estimates are negative and statistically significant, except for the one from column 1.

Further, Table 16 shows the estimation results for a model without additional controls. Again, the main pattern of results remains the same than in Table 1, though the estimate for intra-state conflict is imprecise. Notably, the coefficient for inter-state conflict is statistically significant. In addition, it is clear from Table 17 that dropping unreliable data does not seem to affect our main results much. The estimates are negative and in line with Table 1, though the estimate for intra-state conflict marginally lacks precision (*p*-val.:0.11).

Last, Table 5 presents the estimation results for the first-differenced model in Eq. (6). The aid estimates are all negative and most are statistically significant. The size of the estimates is in line with that of our baseline specification. In other words, we find consistent evidence of the effect of US food aid toward reducing overall and civil conflict incidences even when relying on a first-difference specification. We note, however, that the number of lags seems to matter more with this specification as using deeper lags returns insignificant estimates.[25]

[25] These results are not reported for the sake of space.

### 3.3 Other conflict outcomes

An additional set of estimation results for different conflict outcomes can be found in Appendix Table 18. First, in contrast with NQ, we find that US food aid reduces the onset of intra-state conflicts (Table 18 column 9). The estimate is about − 0.0007 and statistically significant, suggesting the impact of US food aid on the onset of intra-state conflicts is substantial. Second, we study the role of US food aid while considering the scale of conflict (columns 1-8). The conflict measure is disaggregated into small-scale and large-scale conflicts, defined as follows: small-scale conflicts with 25–999 battle deaths and large-scale conflicts with 1,000 or more battle deaths. We find that US food aid decreases the incidence of small-scale conflicts (columns 1-3-5). However, there is no consistent evidence about whether US food aid could impact the incidence of large conflicts (columns 2-4-6-8).

### 3.4 Serial correlation in shifts

As explained in Sect. 2, serial correlation in the shifts could bias our main estimates. We rebuild the SS instrument by using the unpredictable component of the contemporaneous shifts. The latter are extracted by regressing the shifts on a quadratic trend and lagged shifts. Table 6 displays the estimation results of this alternative modeling. The main pattern remains the same. All aid estimates are negative. However, the dynamic estimates lack precision.

### 3.5 Pre/post-Cold War era

Sheppard (2021) suggests the impact of US food aid on conflict is specific to the sample period, especially before/after the Cold War. Table 6 displays the estimation results from the interaction model in Eq. (7).

Put simply, we find no evidence to support his conjecture. Using static and dynamic models, we find that US food aid decreases conflict and intra-state conflict before and after the Cold War. Estimates from the static model suggest that effect of US food aid is larger after the Cold War, while dynamic estimates imply there is no statistical difference between the effect of US food aid before and after the Cold War. As in NQ and CB, we highlight that the identification strategy in Sheppard (2021) is invalid due the incomplete shares (Table 7).

## 4 Conclusions

In this paper, we have revisited the seminal paper of NQ (2014). We explain that their identification strategy is invalid because the shares do not add up to one. The exogeneity of the SS instrument can be restored by adding a sum of shares control interacted with period fixed effects. This solution naturally gives rise to a common factor model representation in NQ. We thus combine their SS instrumentation and

**Table 6** The effect of US food aid on conflict: serial correlation in shifts

| | Any conflict | | Intra-state conflict | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| US wheat aid (1000 MT) | − 0.00092*** | − 0.00015 | − 0.00056* | − 0.00021# |
| | (0.00028) | (0.00020) | (0.00032) | (0.00015) |
| Lagged dependent variable | | 0.76521*** | | 0.72144*** |
| | | (0.04912) | | (0.03369) |
| N | 3,594 | 3,477 | 3,594 | 3,477 |
| Number of countries | 125 | 124 | 125 | 124 |
| Number of instruments | 107 | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 | 4 |
| Number of factors (aid) | 8 | 8 | 8 | 8 |
| Rho | 0.577 | 0.337 | 0.608 | 0.362 |
| J test, *p*-val | 0.451 | 0.433 | 0.439 | 0.434 |
| J-stat | 55.62 | 55.08 | 55.94 | 55.07 |
| BIC | − 820.4 | − 817.4 | − 820.1 | − 817.4 |
| HQIC | − 416.8 | − 416.4 | − 416.5 | − 416.5 |
| Okui, MSE (× 1000) | | 0.877 | | 0.320 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, # $p < 0.15$
Rho is the fraction of the variance of the error term that is explained by the factor component. Country and year fixed effects included. All G2SIV estimates
Sample includes 125 countries between 1971 and 2006. All other controls included (see page 8)

common factor models. The latter control for country-specific nonlinear trends and allows for common shocks to have heterogeneous impacts on conflict across countries.

Using this approach, we find overwhelming evidence that US food aid contributes to reducing the incidence of conflicts in recipient countries. Our estimates suggest that a 1000 MT increase in US wheat aid would decrease the incidence of conflict by 0.1 percentage point. For a country at the sample mean of US wheat aid (2775 MT), a 10% increase in US food aid would result in a 0.3 percentage point decrease in the incidence of conflicts, or 1.5% decrease at the sample mean of conflict incidence.

We also find that US food aid decreases the incidence of civil conflicts, the onset of civil conflicts, as well as the incidence of small-scale conflicts. These results resonate with Mary and Mishra (2020) who find that (overall) humanitarian food aid reduces civil conflict but contrast with NQ and CB. Moreover, our results somewhat undermine the argument that some rebel groups view US food aid differently than food aid from other countries (Koppenberg et al. 2023). We find no effect of US wheat aid on inter-state conflict, whether the conflict is of small or large scale.

More generally, comparing our results to those in the existing and conflicting literature suggests that conclusions about the conflict effects of US food aid are highly sensitive to assumptions about the structure of unobserved shocks (e.g., Nunn and Qian 2014; Christian and Barrett 2023). Under our specifications that allow for latent

**Table 7** The effect of US food aid on conflict: pre/post-Cold War

| | Any conflict | | Intra-state | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Lagged dependent variable | | 0.78231*** | | 0.733549*** |
| | | (0.052621) | | (0.03655) |
| US wheat aid (1000 MT), pre-CW | − 0.00090*** | − 0.00056*** | − 0.000801*** | − 0.000603*** |
| | (0.00025) | (0.00018) | (0.000277) | (0.000192) |
| US wheat aid (1000 MT)*CW | − 0.00115*** | 0.00008 | − 0.000657** | 0.000102 |
| | (0.00040) | (0.00033) | (0.000333) | (0.000258) |
| US wheat aid (1000 MT), post-CW | − 0.00205*** | − 0.000477 | − 0.00146*** | − 0.000501** |
| | (0.00043) | (0.00365) | (0.00036) | (0.000233) |
| N | 3,699 | 3,571 | 3,699 | 3,571 |
| Number of countries | 125 | 124 | 125 | 124 |
| Number of instruments | 107 | 107 | 107 | 107 |
| Rho | 0.585 | 0.324 | 0.616 | 0.349 |
| J test, *p*-val | 0.534 | 0.231 | 0.459 | 0.627 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component
Country and year fixed effects included. All G2SIV estimates. Sample includes 125 countries between 1971 and 2006
All other controls included (see page 8). We use 4 lags and up to 8 common factors

common factors, we obtain negative point estimates, whereas alternative treatments of the error term yield effects that are statistically indistinguishable from zero (Christian and Barrett 2023; Table 10 of this paper), and in some cases even positive (Nunn and Qian 2014; Sheppard 2021).

As we have argued, these modeling choices differ in the strength and plausibility of their identifying assumptions, and we encourage researchers to exercise judgment when selecting among them. At a minimum, empirical work in this context should report robustness checks across a range of specifications that vary in how unobserved common shocks are handled.

Like all approaches that model unobserved common shocks, factor-based methods rely on assumptions about the latent structure of the error term. While we view these assumptions as plausible in our context, we interpret our findings primarily as evidence of specification sensitivity rather than as definitive estimates of a causal effect of food aid on conflict.

Finally, Nunn and Qian (2014) study the period 1971–2006, during which US international food assistance was delivered predominantly as in-kind commodities. Since then, US food aid policy has evolved substantially, most notably with the creation of USAID's Emergency Food Security Program (EFSP) in 2010, which expanded the

use of market-based assistance such as vouchers and direct cash transfers. While in-kind aid has continued to account for a substantial share of US international food assistance until recently (e.g., about 41% as of 2021; Casey and Morgenstern 2021), the growing reliance on market-based modalities could limit the external validity of our estimates for predicting the impacts of future food assistance.

## Appendix

See Tables 8, 9, 10, 11, 12, 13, 14, 15, 16, 17 and 18.

**Table 8** Tests for cross-sectional dependence

| | CD test (Pesaran 2015) H0: errors are weakly CSD | | Exponent CSD (Bailey et al. 2019; 2019) |
|---|---|---|---|
| Variables | Statistic | P-value | Alpha |
| Civil conflict, level | 2.63 | 0.008 | 0.660 |
| *First-difference* | − 0.605 | 0.545 | 0.325 |
| Intra-state conflict, level | 3.510 | 0.000 | 0.582 |
| *First-difference* | − 0.018 | 0.986 | 0.378 |
| Inter-state conflict, level | 0.505 | 0.613 | 0.505 |
| *First-difference* | 0.120 | 0.904 | 0.260 |
| US wheat aid, level | 7.84 | 0.000 | 0.511 |
| *First-difference* | 6.208 | 0.000 | 0.427 |
| SS instrument, level | 302.77 | 0.000 | 0.973 |
| *First-difference* | 332.879 | 0.000 | 0.974 |

0.5 < = alpha < 1 implies strong cross-sectional dependence
We use the Stata command xtcse2 to implement the tests

**Table 9** The effect of US food aid on conflict: replication of NQ and pre-trend tests

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Any conflict | Lagged any conflict | Intra-state | Lagged intra-state | Inter-state | Lagged inter-state |
| US wheat aid | 0.00299*** | 0.00217** | 0.00254*** | 0.00167** | − 0.00044 | − 0.00037 |
| (1000 MT) | (0.00096) | (0.00096) | (0.00088) | (0.00083) | (0.00033) | (0.00036) |
| N | 4,089 | 4,071 | 4,089 | 4,071 | 4,089 | 4,071 |
| KP stat | 12.10 | 11.68 | 12.10 | 11.68 | 12.10 | 11.68 |

Country-clustered robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Country and year fixed effects included. All 2SLS estimates. Sample includes 125 countries between 1971 and 2006. All other controls included (see page 8)
*KP* Kleibergen–Paap statistic

**Table 10** The effect of US food aid changes on conflict changes: implementing CB and pre-trend tests

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Δ Any conflict | Lagged Δ any conflict | Δ Intra-state | Lagged Δ intra-state | Δ Inter-state | Lagged Δ inter-state |
| Δ US wheat aid (1000 MT) | 0.00692 | − 0.00049 | 0.00897 | − 0.00296 | − 0.00183 | 0.00227 |
| | (0.00743) | (0.00468) | (0.00896) | (0.00576) | (0.00301) | (0.00261) |
| N | 3,964 | | 3,964 | | 3,964 | |
| KP stat | 1.51 | | 1.51 | | 1.51 | |

Country-clustered robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Country and year fixed effects included. All 2SLS estimates. Sample includes 125 countries between 1971 and 2006
All other controls included (see page 8). Δ indicates the first-difference of the variable. KP: Kleibergen–Paap statistic

**Table 11** The effect of US food aid on conflict: pre-trend tests

| | (1) | (2) | (3) |
|---|---|---|---|
| | Lagged Any conflict | Lagged Intra-state | Lagged Inter-state |
| US wheat aid (1000 MT) | − 0.00008 | 0.00017 | 0.00000 |
| | (0.00047) | (0.00031) | (0.00006) |
| N | 3,571 | 3,571 | 3,571 |
| Number of countries | 124 | 124 | 124 |
| Number of instruments | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 |
| Number of common factors (aid) | 8 | 8 | 8 |
| Rho | 0.567 | 0.621 | 0.677 |
| J test, *p*-val | 0.528 | 0.399 | 0.968 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component
Country and year fixed effects included. All G2SIV estimates. Sample includes 124 countries between 1971 and 2006. One country is dropped from the sample due to data limitations. All other controls included (see page 8)

**Table 12** The effect of the Bartik instrument on US wheat aid

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | US wheat aid | US wheat aid | US wheat aid | US wheat aid |
| $P_{t-1} * \overline{D}_i$ | 0.00173*** | 0.00062*** | 0.00024*** | 0.00041*** |
| | (0.00013) | (0.00007) | (0.00007) | (0.00005) |
| | [13.57] | [8.70] | [3.71] | [8.20] |
| N | 3,699 | 3,594 | 3,489 | 3,383 |
| Number of instruments | 107 | 108 | 109 | 110 |
| Lags | 4 | 5 | 6 | 7 |
| Number of common factors (instrument) | 8 | 8 | 8 | 8 |
| Rho | 0.618 | 0.904 | 0.910 | 0.910 |
| J test, *p*-val | 0.606 | 0.226 | 0.400 | 0.244 |

Robust standard errors in parentheses. z-stats between square brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component. Country and year fixed effects included. All G2SIV estimates
Factors are extracted from standardized variables. Sample includes 125 countries between 1971 and 2006. All other controls included (see page 8)

**Table 13** The effect of US food aid on conflict: static model, aid exogenous

| | (1) | (2) | (3) |
|---|---|---|---|
| | Any conflict | Intra-state | Inter-state |
| US wheat aid (1000 MT) | − 0.00032*** | − 0.00039*** | − 0.00001 |
| | (0.00011) | (0.00013) | (0.00001) |
| N | 3,647 | 3,647 | 3,647 |
| Number of countries | 125 | 125 | 125 |
| Number of instruments | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 |
| Number of common factors (aid) | 8 | 8 | 8 |
| Rho | 0.572 | 0.617 | 0.693 |
| J-stat | 51.46 | 51.24 | 37.58 |
| J test, *p*-val | 0.611 | 0.619 | 0.965 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component
Country and year fixed effects included. All G2SIV estimates. Sample includes 125 countries between 1971 and 2006. All other controls included (see page 8)

**Table 14** The effect of US food aid on conflict: dynamic model (varying lags)

| | Any conflict | | | | Intra-state | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Lagged dependent variable | 0.80110*** | 0.82583*** | 0.78416*** | 0.78065*** | 0.77394*** | 0.80018*** | 0.72957*** | 0.75217*** |
| | (0.04673) | (0.04679) | (0.04441) | (0.04190) | (0.03291) | (0.03366) | (0.03707) | (0.03986) |
| US wheat aid (1000 MT) | − 0.00030# | − 0.00030* | − 0.00003 | − 0.00029 | − 0.00039** | − 0.00044** | − 0.00003 | − 0.00022 |
| | (0.00019) | (0.00018) | (0.00017) | (0.00022) | (0.00017) | (0.00017) | (0.00009) | (0.00018) |
| N | 3,477 | 3,323 | 3.199 | 3,075 | 3,477 | 3,323 | 3.199 | 3,075 |
| Number of countries | 124 | 124 | 124 | 124 | 124 | 124 | 124 | 124 |
| Number of instruments | 108 | 109 | 110 | 111 | 108 | 109 | 110 | 111 |
| Lags | 5 | 6 | 7 | 8 | 5 | 6 | 7 | 8 |
| Rho | 0.340 | 0.344 | 0.346 | 0.352 | 0.369 | 0.379 | 0.384 | 0.396 |
| J test, *p*-val | 0.542 | 0.471 | 0.548 | 0.605 | 0.455 | 0.604 | 0.699 | 0.600 |
| J-stat | 53.24 | 56.11 | 55.06 | 54.53 | 55.51 | 52.62 | 50.98 | 54.66 |
| BIC | − 826.4 | − 827.7 | − 832.7 | − 836.9 | − 824.2 | − 831.2 | − 836.8 | − 836.8 |
| HQIC | − 422.5 | − 423 | − 427.3 | − 431.1 | − 420.2 | − 426.5 | − 431.4 | − 431 |
| Okui, MSE (× 1000) | 1.43 | 2.29 | 1.42 | 1.51 | 0.582 | 0.881 | 0.414 | 0.560 |
| CD stat, residuals | 5.124 | 8.337 | 4.566 | 5.735 | 8.214 | 9.437 | 6.614 | 9.082 |
| Exponent CSD | 0.320 | 0.338 | 0.311 | 0.327 | 0.268 | 0.283 | 0.286 | 0.293 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, # $p < 0.15$
Rho is the fraction of the variance of the error term that is explained by the factor component
Country and year fixed effects included. All G2SIV estimates. Sample includes 124 countries between 1971 and 2006
One country is dropped from the sample due to data limitations. All other controls included (see page 8)
Number of factors is up to 8

**Table 15** The effect of US food aid on conflict: dynamic model (varying factors)

| | Any conflict | | | | Intra-state | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Lagged dependent variable | 0.77150*** | 0.74393*** | 0.77155*** | 0.77100*** | 0.74946*** | 0.75522*** | 0.72881*** | 0.76069*** |
| | (0.04450) | (0.04278) | (0.04622) | (0.04561) | (0.03580) | (0.03029) | (0.03300) | (0.03303) |
| US wheat aid (1000 MT) | − 0.00022 | − 0.00052* | − 0.00045# | − 0.00063** | − 0.00075*** | − 0.00071*** | − 0.00072*** | − 0.00061*** |
| | (0.00027) | (0.00030) | (0.00030) | (0.00028) | (0.00022) | (0.00021) | (0.00017) | (0.00019) |
| N | 3,571 | 3,571 | 3,571 | 3,571 | 3,571 | 3,571 | 3,571 | 3,571 |
| Number of countries | 124 | 124 | 124 | 124 | 124 | 124 | 124 | 124 |
| Number of instruments | 107 | 107 | 107 | 107 | 107 | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| Number of common factors (aid) | 6 | 7 | 9 | 10 | 6 | 7 | 9 | 10 |
| Rho | 0.317 | 0.320 | 0.320 | 0.327 | 0.350 | 0.350 | 0.349 | 0.352 |
| J test, *p*-val | 0.410 | 0.505 | 0.506 | 0.442 | 0.652 | 0.707 | 0.749 | 0.603 |
| J-stat | 55.72 | 53.20 | 53.18 | 54.85 | 49.40 | 47.92 | 46.70 | 50.68 |
| BIC | − 819.6 | − 822.1 | − 822.1 | − 820.5 | − 825.9 | − 827.4 | − 828.6 | − 824.6 |
| HQIC | − 416.5 | − 419.1 | − 419.1 | − 417.4 | − 422.9 | − 424.3 | − 425.6 | − 421.6 |

**Table 15** (continued)

| | Any conflict | | | | Intra-state | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Okui, MSE (× 1000) | 0.937 | 0.765 | 0.993 | 1.063 | 0.515 | 0.529 | 0.416 | 0.528 |
| CD stat, residuals | 5.859 | 7.854 | 7.767 | 7.362 | 6.434 | 7.600 | 6.908 | 7.591 |
| Exponent CSD | 0.418 | 0.485 | 0.454 | 0.466 | 0.289 | 0.293 | 0.289 | 0.288 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, # $p < 0.15$
Rho is the fraction of the variance of the error term that is explained by the factor component
Country and year fixed effects included. All G2SIV estimates. Sample includes 124 countries between 1971 and 2006
One country is dropped from the sample due to data limitations. All other controls included (see page 8)
$p$-val for CD test is 0.00 for all columns

**Table 16** The effect of US food aid on conflict: static model (no additional covariates)

| | (1) | (2) | (3) |
|---|---|---|---|
| | Any conflict | Intra-state | Inter-state |
| US wheat aid (1000 MT) | − 0.00632*** | − 0.00128 | − 0.01033*** |
| | (0.00141) | (0.00166) | (0.00160) |
| N | 3,699 | 3,699 | 3,699 |
| Number of countries | 125 | 125 | 125 |
| Number of instruments | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 |
| Number of common factors (aid) | 8 | 8 | 8 |
| Rho | 0.810 | 0.632 | 0.887 |
| J test, *p*-val | 0.209 | 0.111 | 0.392 |
| BIC | − 35.21 | − 33.57 | − 36.98 |
| HQIC | − 16.24 | − 14.60 | − 18.01 |
| CD stat, residuals | 10.29 | 7.469 | 12.73 |
| Exponent CSD | 0.647 | 0.596 | 0.572 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Rho is the fraction of the variance of the error term that is explained by the factor component
Country and year fixed effects included. All G2SIV estimates
Sample includes 125 countries between 1971 and 2006. No other control included

**Table 17** The effect of US food aid on conflict: excluding data 1971–1974

| | (1) | (2) | (3) |
|---|---|---|---|
| | Any conflict | Intra-state | Inter-state |
| US wheat aid (1000 MT) | − 0.00075** | − 0.00048# | − 0.00000 |
| | (0.00031) | (0.00030) | (0.00005) |
| N | 3,383 | 3,383 | 3,383 |
| Number of countries | 125 | 125 | 125 |
| Number of instruments | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 |
| Number of common factors (aid) | 8 | 8 | 8 |
| Rho | 0.587 | 0.610 | 0.700 |
| J-stat | 49.07 | 58.34 | 36.79 |
| J test, *p*-val | 0.699 | 0.354 | 0.972 |
| BIC | − 820.5 | − 811.2 | − 832.8 |
| HQIC | − 421.7 | − 412.4 | − 434 |
| CD stat, residuals | 9.052 | 11.32 | 77.34 |
| CD test, residuals, *p*-val | 0.000 | 0.000 | 0.000 |
| Exponent CSD | 0.610 | 0.557 | 0.434 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, # $p < 0.15$
Rho is the fraction of the variance of the error term that is explained by the factor component
95% RI confidence intervals calculated over 1000 counterfactuals. Country and year fixed effects included. All G2SIV estimates
Sample includes 125 countries between 1975 and 2006. All other controls included (see page 8)

**Table 18** The effect of US food aid on alternative conflict outcomes

| | Any conflict | | | | Intra-state | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Small | Large | Small | Large | Small | Large | Small | Large | Onset |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| US wheat aid (1000 MT) | − 0.00050* | − 0.00006 | − 0.00067*** | − 0.00004 | − 0.00097*** | − 0.00016 | − 0.00020 | − 0.00015# | − 0.00065*** |
| | (0.00028) | (0.00017) | (0.00023) | (0.00007) | (0.00029) | (0.00011) | (0.00015) | (0.00009) | (0.00013) |
| Lagged dependent variable | | | 0.57363*** | 0.70304*** | | | 0.67222*** | 0.88522*** | |
| | | | (0.04984) | (0.01966) | | | (0.04097) | (0.01834) | |
| N | 3,699 | 3,699 | 3,571 | 3,571 | 3,699 | 3,699 | 3,571 | 3,571 | 3,699 |
| Number of countries | 125 | 125 | 124 | 124 | 125 | 125 | 124 | 124 | 125 |
| Number of instruments | 107 | 107 | 107 | 107 | 107 | 107 | 107 | 107 | 107 |
| Lags | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| Number of common factors (aid) | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, # $p < 0.15$
Rho is the fraction of the variance of the error term that is explained by the factor component
Country and year fixed effects included
All G2SIV estimates. Sample includes 125 countries between 1971 and 2006
All other controls included (see page 8)

**Data availability** Data are publicly available at the following link: https://www.openicpsr.org/openicpsr/project/112825/version/V1/view.

## Declarations

**Conflicts of Interest** The author declares no conflict of interest.

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